

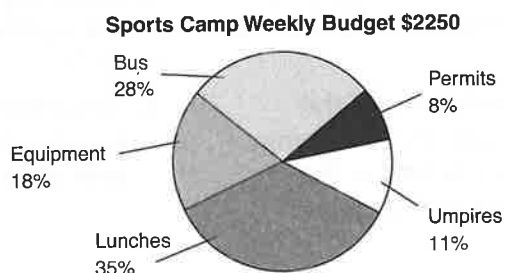
Lesson 3: Circle Graphs

Key Ideas

- In a circle graph, the circle represents a whole amount, and the sections represent parts of the whole.
- Sections are usually labeled with percents, but they can show amounts.
- The size of each section of a circle graph is equal to the fraction or percent it represents.

A **circle graph** is used to show how a whole amount is broken into parts. The sections of a circle graph are often labeled with percents. The size of each section corresponds to the fraction it represents. For example, a section labeled 25% is $\frac{1}{4}$ of the circle.

Example 1: A graph below shows how a children's sports camp spends its weekly budget.



How much does the sports camp spend on lunches each week?

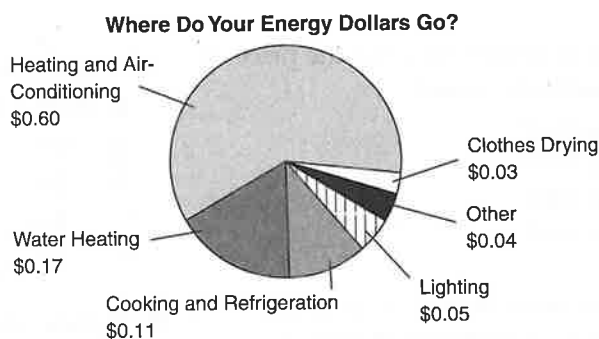
1. **Analyze the graph.** According to the heading, the entire circle represents the camp's weekly budget of \$2250. Find the section labeled "lunches." According to the section label, lunches make up 35% of the weekly budget.
2. **Use the data.** To find the amount spent on lunches, find 35% of \$2250:
 $\$2250 \times 0.35 = \787.50 .

A circle graph may also be labeled using fractions or decimals. One common kind of circle graph labels each section in cents to show how a dollar is used.

GED® TEST TIP

Since a circle graph represents a whole, or 100%, amounts can easily be converted to percents. For example, \$0.05 on a circle graph can also represent 5%.

Example 2: According to the graph, what percent of the average energy bill is spent on drying clothes, lighting, and heating water?



1. **Analyze the graph.** The entire circle represents \$1. The amounts in the sections mentioned in the problem are \$0.03, \$0.05, and \$0.17.
2. **Use the data.** Add the amounts: $\$0.03 + \$0.05 + \$0.17 = \0.25 . Since \$0.25 is 25% of a dollar, 25% of an average bill is spent on these items.

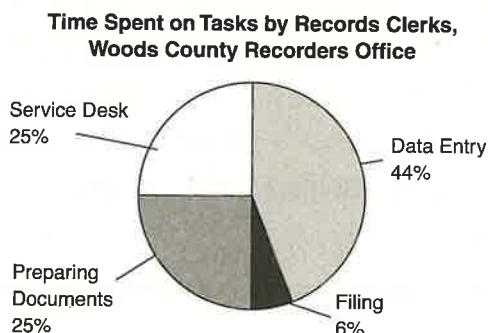
Practice 3

A. For questions 1 through 3, use the Sports Camp Weekly Budget circle graph on page 286. For questions 4 through 6, use the Where Do Your Energy Dollars Go? circle graph on page 286.

1. What percent of the total sports camp budget is spent on equipment and umpires?
2. What fraction of the sports camp budget is spent on permits?
3. What amount does the camp spend each week on busing?
4. A family's energy bill is \$180. Assuming the family's energy use is typical, how much did the family spend on water heating?
5. Which section is greater than 50 percent of an energy dollar?
6. Which energy cost is about $\frac{1}{10}$ of the energy dollar?

B. Choose the one best answer to each question.

Questions 7 and 8 refer to the following graph.



7. During a 40-hour workweek, how many hours does a records clerk spend preparing documents?
8. What percent of a records clerk's time is spent on tasks other than data entry?

Questions 9 and 10 refer to the following graph.

The employees of National Bank are given the following graph to explain how their retirement fund is invested.



9. What percent of each retirement dollar is invested in real estate and the stock market?
10. Steve contributes \$120 of each paycheck to his National Bank retirement fund. How much of each contribution is invested in public bonds?

Answers and explanations begin on page 668.

Lesson 4: Measures of Central Tendency

Using a Frequency Table

A **frequency table** shows how often an item appears in a data set. The data is in the form of tally marks next to a list of items.

Example 1: The sales manager at Montana Motors asked her sales staff to keep a record of the color of the cars that were chosen for test-drives in one month. Then she combined the data to make the frequency table shown below.

Montana Motors—Car Color Preferences	
white	
black	
red	
green	
silver	
other	

What was the ratio of black cars driven to silver cars driven?

- Count the tally marks. There are 27 marks for black and 9 for silver.
- Write the ratio and reduce to lowest terms. $\frac{27}{9} = \frac{3}{1}$

The ratio of black to silver is **3 to 1**. You can also say that the black cars are 3 times as popular as the silver cars.

Numerical data is often grouped in intervals. The table below shows data grouped in intervals of 18 to 24, 25 to 40, and so on. This way of presenting data is called a **grouped frequency table**.

Example 2: The table below shows the ages of the customers at Louise's Diner for a four-day period. What percent of the customers were from 25 to 40 years old?

Louise's Diner Customers by Age Group, February 19–22	
under 18	
18–24	
25–40	
41–55	
over 55	

- Find the data you need. There are 28 marks for the 25–40 age group. Add the tally marks for all age groups to find the total number of customers for the three-day period: $6 + 18 + 28 + 10 + 8 = 70$.
- Find the percent. The base is 70, the total number of customers. The part is 28, the number of customers in the desired age group. Solve for the rate: $\frac{28}{70} = \frac{4}{10} = 0.4 = 40\%$.

Key Ideas

- A frequency table is used to show how often data occurs.
- Tally marks are made next to a list to record the data.
- Numerical data is often grouped in intervals.

GED® TEST TIP

If several problems refer to a frequency table, you may want to count the tally marks for each line and record the numbers on the wipe-off board provided at the testing center. Then use the numbers to solve all related problems.

Practice 4.1

- A. For questions 1 through 3, use the frequency table from Montana Motors on page 288. For questions 4 through 6, use the frequency table from Louise's Diner on page 288.

1. What was the total number of test drives of black and white cars combined?
2. How many more drivers chose red than silver cars?
3. What was the ratio of red cars to white cars chosen for test-drives?
4. What is the ratio of customers under 18 to those over 55?
5. What was the total number of customers from 18 to 40 years of age?
6. What percent of the total customers were from 41 to 55 years of age? (Round your answer to the nearest whole percent.)

- B. Choose the one best answer to each question.

Questions 7 and 8 refer to the following information.

The frequency table shows the reasons customers gave for returning clothing merchandise to a store.

Reason	Number
Wrong size	
Unwanted gift	
Found flaw after purchase	
Changed mind	

7. What is the ratio of customers saying the clothes were the wrong size to all the returns represented?

- A. $\frac{4}{15}$
- B. $\frac{4}{11}$
- C. $\frac{4}{7}$
- D. $\frac{4}{5}$

8. Approximately what percent of the customers who returned clothing said that the clothes were an unwanted gift?

- A. 20%
- B. 25%
- C. 45%
- D. 50%

Questions 9 and 10 refer to the following information.

A personnel office gives typing tests to people applying for a job. The test shows how many words per minute (wpm) a job applicant can enter correctly. After testing 90 applicants, the manager made the following table.

Typing Speed	Number
Under 30 wpm	
30–45 wpm	
46–60 wpm	
Over 60 wpm	

9. What percent of the applicants had a speed of under 30 wpm?

- A. 14%
- B. 25%
- C. 28%
- D. 39%

10. What is the ratio of applicants who could type at a speed above 45 wpm to those who could type at a speed of 45 wpm or less?

- A. 1:3
- B. 1:2
- C. 2:3
- D. 6:5

Answers and explanations begin on page 668.

Mean, Median, and Mode

Suppose you were asked how much money you usually spend on groceries in a week. Some weeks, you may spend a great deal; other weeks, much less. You would probably choose an amount in the middle to represent what you typically spend. This middle value is called an **average**, or **measure of central tendency**.

The most common type of average is the **mean**, or the arithmetic average.

Example 3: In five football games, a team scored 14, 21, 3, 20, and 10 points. What is the mean, or average, score per game?

1. Add the values. $14 + 21 + 3 + 20 + 10 = 68$
2. Divide by the number of items in the data set. $68 \div 5 = \mathbf{13.6 \text{ points per game}}$

Although it is impossible for a football team to score 13.6 points in a game, the number represents the center of the scores from the five games.

A calculator is useful for finding the mean. Do the calculations in two steps. Enter the addition operations. Then, if you are using the TI-30XS MultiView™, press *enter* to find the sum. Then key in the division operation. Try Example 1 above with a calculator.

Another measure of average is the median. The **median** is the middle value in a set of data.

Example 4: During a 7-hour period, a bookstore recorded the following numbers of sales. Find the median number of sales.

Hour 1	Hour 2	Hour 3	Hour 4	Hour 5	Hour 6	Hour 7
43	28	24	36	32	37	48

1. Arrange the values by size. 24, 28, 32, 36, 37, 43, 48
2. Find the middle number. 24, 28, 32, **36**, 37, 43, 48

If there is an even number of values, the median is the mean of the two middle values.

Example 5: Robert has the following test scores in his math class: 90, 72, 88, 94, 91, and 80. What is the median score?

1. Arrange the values by size and find the middle. 72, 80, **88, 90**, 91, 94
2. Find the mean of the two middle values. The median score is **89**.
Add: $88 + 90 = 178$
Divide by 2: $178 \div 2 = 89$

The **mode** is the value that occurs most often in a set of data. A set of data could have more than one mode if several items occur the same number of times. If each item of data occurs only once, there is no mode.

Example 6: Six weather stations recorded the following temperatures at 3:00 p.m.: 45°, 44°, 45°, 47°, 46°, and 45°. What is the mode of the data?

The temperature 45° occurs most often (3 times). The mode is **45°**.

Practice 4.2

A. For each data set, find the mean, median, and mode. Round calculations to the nearest hundredth or cent. You MAY use a calculator.

- Golf scores for 7 rounds:
76, 82, 75, 87, 80, 82, and 79
- Sales totals for 6 weeks:
\$5,624; \$10,380; \$8,102; \$6,494; \$12,008;
and \$8,315
- Cost of lunch for 8 days:
\$4.50, \$5.25, \$4.50, \$3.75, \$4.50, \$5.25,
\$6.10, and \$4.25
- Miles driven per day for 5 days:
330, 286, 342, 300, and 287
- Grocery bills for 4 weeks:
\$97.48, \$106.13, \$110.98, and \$92.74
- Scores on 7 quizzes:
90, 72, 86, 100, 88, 78, and 88
- High temperatures for 10 days:
96°, 103°, 98°, 101°, 98°, 100°, 100°, 97°,
98°, and 100°
- Inches of rainfall over 3-day period:
2.5, 1.8, and 1.4
- Attendance figures at a play:
305, 294, 328, 296, 305, 315, and 292
- Hours worked per week for 5 weeks:
36, 40, 38, 40, and 40

B. Choose the one best answer to each question. You MAY use your calculator.

Questions 11 and 12 refer to the following information.

Homes Sold in Fairfield Heights in June

Home	Asking Price	Selling Price
#1	\$124,600	\$116,500
#2	\$132,400	\$124,800
#3	\$118,900	\$116,500
#4	\$98,500	\$103,600
#5	\$105,800	\$109,000
#6	\$122,400	\$118,400

- What was the mean asking price of the homes sold in Fairfield Heights in June?
A. \$117,100
B. \$116,500
C. \$115,450
D. \$114,800
- What was the median selling price of the homes sold in Fairfield Heights in June?
A. \$112,750
B. \$114,800
C. \$116,500
D. \$117,450
- The numbers of patients enrolled at four health clinics are 790, 1150, 662, and 805. Which expression could be used to find the mean number of patients per clinic?
A. $\frac{790 + 1150 + 662 + 805}{4}$
B. $790 + 1150 + 662 + 805$
C. $\frac{662 + 1150}{2}$
D. $(790 + 1150 + 662 + 805) \div 2$
- What is the median value of \$268, \$1258, \$654, \$1258, \$900, \$1558, and \$852?
A. \$1258
B. \$964
C. \$900
D. \$852
- What is the mode of the following points scored: 14, 17, 14, 12, 13, 15, 22, and 11?
A. 13.5
B. 14
C. 14.75
D. 16.5

Answers and explanations begin on page 668.

Lesson 5: Line Plots

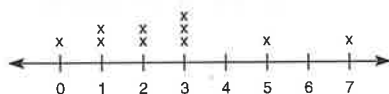
Key Ideas

- A line plot is a graph that shows the frequency of data along a number line.
- Based on frequency tables, line plots consist of a horizontal scale and points or x 's placed over corresponding numbers.
- Line plots help us to interpret the distribution of data.

Example 1: The student health services department at a university surveyed several students and asked them how many times per week they visited the school's gym. The following frequency table shows the results for several students.

Weekly Gym Visits	Number of Students
0	I
1	II
2	II
3	III
4	0
5	I
6	0
7	I

Next, the health services department decided to create a line plot in order to see the distribution of gym visits. Each x represents a student.



You can see from this line plot that there is an uneven **distribution**, or arrangement, of students across the numbers of possible gym visits. Three visits per week has the **highest frequency**, meaning that more students go to the gym three times per week than any other number of visits. You can also see that the data **range** extends from zero visits per week to seven visits per week—no student visits the gym more often than that. The data point showing the one student who visits the gym seven times per week is an **outlier**—that is, a data point that is distant from where most of the data is clustered.

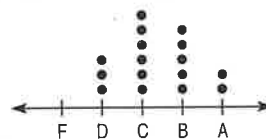
ON THE GED® TEST

One type of question may ask you to analyze data on a line plot. Another type of question could ask you to use a frequency table to place an element on a line plot.

The *Mathematical Reasoning Test* may ask you to identify highest frequency, range, and outliers, or it may give you some data and ask you to place x 's or dots on a number line to create a line plot.

Use the following example to study these concepts further.

Example 2: Students in Ms. Jones's class took a math test. Their scores on the test are displayed on the line plot below. Each dot represents a student. Which grade displayed the highest frequency?



More students received a C than received any other grade. Therefore, grade C has the highest frequency of any grade on the line plot.

The range of grades on the math test was from D (the lowest grade received by any student) to A (the highest grade received by any student).